

Code-along 03

FirstName LastName

Setup

Packages

Load the standard packages.

```
library(tidyverse)
library(haven) # not core tidyverse
library(gssr)
library(gssrdoc) # load the GSS codebook
library(summarytools)
```

Load your data & codebook

```
# Get the data all survey years
data(gss_all)
```

Use the `gss_which_years()` function to identify the survey years a variable appeared.

```
gss_which_years(gss_all, meovrwrk)
```

```
# A tibble: 35 x 2
  year      meovrwrk
  <dbl> <lgl>
1 1972     FALSE
2 1973     FALSE
3 1974     FALSE
4 1975     FALSE
```

```
5 1976      FALSE
6 1977      FALSE
7 1978      FALSE
8 1980      FALSE
9 1982      FALSE
10 1983     FALSE
# i 25 more rows
```

Which survey years did the variable `childs` appear?

```
# REPLACE THE BLANK LINE WITH THE NAME OF THE VARIABLE

gss_which_years(gss_all, _____)
```

Data Management I

The pipe `|>`

The pipe operator passes what comes before it into the function that comes after it as the first argument in that function.

```
sum(1, 2)
```

```
[1] 3
```

```
1 |>
  sum(2)
```

```
[1] 3
```

`dplyr()` in action

Create a frequency table for `childs` variable.

```

gss_all$chlds <- zap_missing(gss_all$chlds)
gss_all$chlds <- as_factor(gss_all$chlds)
gss_all$chlds <- droplevels(gss_all$chlds)

gss_all |>
  freq(chlds, report.nas = FALSE) |>
  tb()

```

```

# A tibble: 9 x 4
  chlds    freq    pct pct_cum
  <fct>    <dbl> <dbl>   <dbl>
1 0      20956 27.8     27.8
2 1      11979 15.9     43.7
3 2      18989 25.2     68.9
4 3      11671 15.5     84.3
5 4       5996  7.95    92.3
6 5       2669  3.54    95.8
7 6       1381  1.83    97.7
8 7        734  0.973   98.6
9 8 or more 1032  1.37   100

```

Use a `|>` & `gss_which_years()` to find out which survey years the variable `agekdbrn` appears.

```
# ADD YOUR CODE HERE
```

`filter()` & `drop_na()`

Use `filter()` to keep rows that meet a condition.
Use `drop_na()` to remove rows with missing (NA) values.

```

gss_all |>
  filter(year == 2022) |>
  drop_na(sex, agekdbrn) |>
  count(sex)

```

```

# A tibble: 2 x 2
  sex      n
  <dbl+lbl> <int>
1 1 [male] 1031
2 2 [female] 1363

```

Create a dataframe with the variables `year`, `sex`, `hrs1`, and `childs` for only the 2024 respondents without missing data for `hrs1` and `childs`.

```
gss_all |>
  filter(year == 2022) # CHANGE THE YEAR TO 2024 & ADD A PIPE OPERATOR
  drop_na() |> # ADD THE VARIABLES INSIDE drop_na()
  count(sex)
```

Summary Statistics

Median & Mean w base R

```
# Using base R
median(gss_all$agekdbnr, na.rm=TRUE)
```

```
[1] 23
```

Note: `na.rm` is a logical evaluating to `TRUE` or `FALSE` indicating whether NA values should be stripped before the computation proceeds.

```
mean(gss_all$agekdbnr, na.rm=TRUE)
```

```
[1] 24.14491
```

```
# show me summary statistics
summary(gss_all$agekdbnr)
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.	NA's
9.00	20.00	23.00	24.14	28.00	97.00	46832

Find the mean and median for the variable: `hrs1`.

```
# REPLACE THE BLANK LINES WITH THE DATA FRAME NAME & VARIABLE NAME
median(_____, na.rm=TRUE)
mean(_____, na.rm=TRUE)

# USE summary() TO SHOW SUMMARY STATISTICS
```

summarise()

Use functions inside **summarise()** to create summary variables.

```
gss_all |>
  drop_na(agekdbnr) |>
  summarise(
    avg = mean(agekdbnr)
  )
```

```
# A tibble: 1 x 1
  avg
<dbl>
1  24.1
```

Median & Mean with dplyr

Use dplyr grammar to find the frequency, mean, and median.

Add lots of summary variables using functions inside **summarise()**.

1. Use **n()** function to create a count variable (no variable in parentheses)
2. Use **mean()** function on a variable
3. Use **median()** function on a variable

```
gss_all |>
  drop_na(agekdbnr) |>
  summarise(
    freq = n(),
    avg = mean(agekdbnr),
    med = median(agekdbnr)
  )
```

```
# A tibble: 1 x 3
  freq  avg  med
<int> <dbl> <dbl>
1 28867  24.1   23
```

Use dplyr grammar to find the frequency, mean, and median for the variable: **hrs1**.

```
gss_all |>
  drop_na(hrs1) |>
  summarise(
    # ADD TO THE PIPE HERE WITH YOUR CODE HERE
  )
```

group_by() and summarize()

A common pipeline is `group_by()` and then `summarize()` to calculate summary statistics for each group:

Use `group_by()` to organize your data into groups based on one or more variables.
Use `summarize()` to compute statistics like total, mean, or median for each group.

1. Start with the `gss_all` data frame:
2. Remove any observations with missing data for our key variables
3. Do the next steps separately for each group in the variable
4. Creates a new data frame with one row for each combination of grouping variables

```
gss_all |>
  drop_na(sex, agekdbn) |>
  group_by(sex) |>
  summarise(avg = mean(agekdbn))
```

```
# A tibble: 2 x 2
  sex      avg
<dbl> <dbl>
1 1 [male]  25.7
2 2 [female] 23.0
```

Compare the average and median work hours (`hrs`) for U.S. men and women.

```
1 gss_all |>
2   drop_na(sex, hrs1) |>
3   # ADD TO THE PIPE HERE WITH YOUR CODE HERE
```

summarize() + variability measures

```
gss_all |>
  drop_na(agekdbnr, sex) |>
  group_by(as_factor(sex)) |>
  summarise(
    freq = n(),
    min = min(agekdbnr),
    median = median(agekdbnr),
    max = max(agekdbnr),
    mean = round(mean(agekdbnr), digits = 2),
    sd = sd(agekdbnr)
  )
```

```
# A tibble: 2 x 7
  `as_factor(sex)`   freq min      median max      mean    sd
  <fct>             <int> <dbl+lbl> <dbl> <dbl+lbl> <dbl> <dbl>
1 male              11790 9         25 65      25.7  5.85
2 female            17009 9         22 57      23.0  5.30
```

Think Like a Statistician

How did the mean and standard deviation of weekly work hours differ between men and women in 2024, and what do these differences suggest about gender-based variability in labor patterns.

How do we find out?

```
# ADD YOUR CODE HERE
```

Your Data Take

What's your conclusion to our initial research question?